

Hallucination, Sycophancy, and Human Cognitive Bias May Share the Same Structure

— A Structural Hypothesis Using SCOPE —

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Chapter 0: Hypothesis and Scope

The Sephirotic names used in this paper are not employed in a religious or mystical sense. In SCOPE, they are borrowed as names, or labels, for distinguishing cognitive structures. The name SCOPE (Sephirotic Cognitive Process Engine) is also based on this borrowing.

This paper does not treat humans and LLMs as identical entities. Their internal structures differ. What this paper examines is the comparison and analysis of their cognitive structures within the same coordinate system, using observable outputs as a shared point of reference.

In this paper, SCOPE is used as a coordinate system. The contents presented here do not prove the causes of hallucination, sycophancy, or human cognitive biases. Rather, they are treated as structural hypotheses in which observed outputs are provisionally mapped onto SCOPE. For the design principles of SCOPE, see the SCOPE integration paper, SCOPE — How the Cognitive Coordinate System Works.

The phrase “may share the same structure” in the title does not mean that hallucination, sycophancy, and human cognitive biases are themselves identical phenomena. Nor does it mean that the states or contributions of all cognitive functions are completely identical. The meaning of “same” as used in this paper is defined in Chapter 1.

This paper compares not phenomenon labels themselves, but the cognitive functions hypothesized to operate behind them. The term “function” is also defined in Chapter 1.

The scope of this paper is limited to aspects of hallucination, sycophancy, and human cognitive biases that are considered comparable from a functional perspective. It does not aim to explain the full set of formation factors underlying each of these phenomena. Nor does it aim to comprehensively cover cognitive biases in general or sycophancy in general. Other phenomena may also be describable using similar structures, but they fall outside the scope of this paper. The range of sycophancy addressed here is defined in Chapter 1.

This paper does not address the formation mechanisms or measurement issues of the Dunning–Kruger effect itself. As a human-side comparison target, this paper focuses on

cognitive structures in which an inflated self-evaluation based on weak or incorrect grounds is formed and the direction of that inflated self-evaluation remains maintained even after information or feedback that would weaken it is introduced. This temporally maintained structure is distinguished from the Dunning–Kruger effect itself and is referred to in this paper as “Dunning–Kruger-type self-evaluation maintenance.”

Some of the structures proposed in this paper may appear to overlap with existing theories or concepts. Their relationships to those theories and concepts fall outside the scope of this paper and are not addressed here.

Definitions of the individual layers used in this paper are kept to the minimum necessary. For detailed definitions and operational methods of each layer, see the preceding SCOPE integration paper, SCOPE — How the Cognitive Coordinate System Works. That paper provides the foundational description of the cognitive coordinate system used here.

Note

By providing the SCOPE integration paper, SCOPE — How the Cognitive Coordinate System Works, to an LLM, it may be possible to use the model as a dialogue partner that assists with SCOPE-based analysis and explanations of the individual layers. This behavior is currently based on N=1 observation and is not guaranteed to reproduce across all LLMs or usage environments. Results may also differ depending on the user and observation conditions.

Chapter 1: Definitions

Chapter 0 presented the purpose and scope of this paper. This chapter defines the major terms used throughout the paper.

Some of the terms used in this paper are defined differently from their general meanings. These definitions have been organized on the basis of the cognitive framework developed by the author. Therefore, even terms that are not explicitly defined here may still differ from their ordinary usage. Readers are asked to prioritize the definitions provided in this paper.

The major terms used in this paper are defined below.

“Same” (Equivalent)

In general usage, “same” refers to objects or properties being identical or matching.

In this paper, “same” does not mean that the objects as a whole are completely identical.

When a single type of function is shared, this paper treats it as the same function. However, for two cases to be described as having the “same structure,” sharing a single function is not sufficient. There must be correspondence between the roles of multiple functions being compared and the relationships among those functions. In this paper, the range within which such correspondence is identified is referred to as the “same structure.”

Therefore, even when physical implementation, cognitive content, the states of individual functions, or the degree of their contribution differ, the corresponding range may still be treated as the same structure when correspondence can be identified among multiple functions and their relationships.

For example, smoke signals, fax, email, and smartphones differ in medium, speed, amount of information, and implementation. However, when focusing on the function of conveying information to another person, they share the function of transmission. The fact that this single function is shared does not mean that these communication methods as a whole are treated as having the same structure.

Likewise, humans and LLMs differ in implementation. Hallucination, sycophancy, and human cognitive bias are also different phenomena. This paper does not compare these phenomena as if they were identical in their entirety. Rather, it examines the possibility that correspondence may be identified between the roles of multiple cognitive functions provisionally mapped behind each phenomenon and the relationships among those functions.

Function

In general usage, a “function” refers to the role or operation performed by something.

In this paper, “function” refers to the operation performed by a target. Even when implementations or forms differ, if the operation being performed is the same, this paper treats them as the same function.

This paper compares functions that may be shared by hallucination, sycophancy, and some forms of human cognitive bias. The specific content of these functions is examined from Chapter 2 onward.

Sycophancy

In general usage, “sycophancy” refers to aligning with another person’s opinions or expectations.

In this paper, sycophancy refers to cases in which a user presents a particular judgment, premise, or direction, and an output aligned with that direction is generated even though evidence that does not support it, opposing information, or information indicating uncertainty has been presented or is externally verifiable, and such information is not sufficiently reflected in the output.

Therefore, the mere fact that an output agrees with the user’s judgment is not sufficient for it to be treated as sycophancy in this paper. A response in the same direction may also be justified when the user’s position is consistent with external facts or sufficiently supported by evidence.

This paper does not attempt to define or explain all factors underlying sycophancy as a phenomenon. Rather, when an output of the kind described above is observed, this paper examines what kind of cognitive structure may be provisionally mapped behind it and compares its similarities and differences with hallucination and human cognitive bias.

Understanding

In general usage, “understanding” refers to grasping the meaning or content of something.

In this paper, “understanding” is defined as constructing an internal model of a target, using that model to infer aspects such as the target’s meaning, state, or intent, and being able to use those in inference or response generation.

Event

In general usage, an “event” refers to something personally experienced.

In this paper, an “event” refers to an occurrence that is actually encountered. The presence or absence of a body is irrelevant, and generation by an LLM is also treated as an event.

Learning

In general usage, “learning” refers to acquiring knowledge or skills.

In this paper, “learning” refers to a process or state in which events, knowledge, and experience are updated. This also includes intermediate stages of updating and states in which updating remains possible.

Experience

In general usage, “experience” refers to knowledge or skills acquired through seeing, hearing, or doing something, or to their accumulation.

In this paper, “experience” refers to an internal structure formed and accumulated through learning, knowledge, and events, which continues to influence subsequent cognition and output.

Knowledge

In general usage, “knowledge” refers to information or understanding acquired through learning or experience.

In this paper, “knowledge” refers to information formed through learning, events, and experience and retained in a compressed form.

Relationship Among Event, Learning, Experience, and Knowledge

In this paper, event, learning, experience, and knowledge are not treated as a one-way process beginning from any single element. These four elements are treated as a cyclical

structure in which they mutually influence one another and each contributes to the updating of the others.

Empathy

In general usage, “empathy” refers to sharing another person’s emotions.

In this paper, “empathy” is defined as inferring another person’s state and reflecting the result of that inference in the output.

Coherence

In general usage, “coherence” refers to a state in which there is little contradiction in logic, writing, or thought.

In this paper, “coherence” refers to organizing and maintaining multiple cognitions related to a cognitive agent in a direction with minimal internal contradiction.

In humans, values, beliefs, worldviews, and self-image may be involved in this coherence.

In LLMs, system policies, conversational roles, policies or directions maintained during response generation, and consistency with preceding outputs may function as elements corresponding to coherence.

In SCOPE, this function is assigned to L3 (Self-Consistency).

However, human self-image or beliefs and an LLM’s system policies or conversational consistency are not treated as the same internal state. This paper considers only the possibility that they correspond in functional position as mechanisms for organizing and maintaining coherence.

The coherence described here refers to coherence established within the cognitive agent itself (internal coherence). Therefore, internal coherence may still be present even when the cognition appears contradictory from an external perspective or does not correspond to reality. Coherence therefore does not guarantee correspondence with reality or logical correctness.

Chapter 2: The Premises for Capturing Hallucination through SCOPE

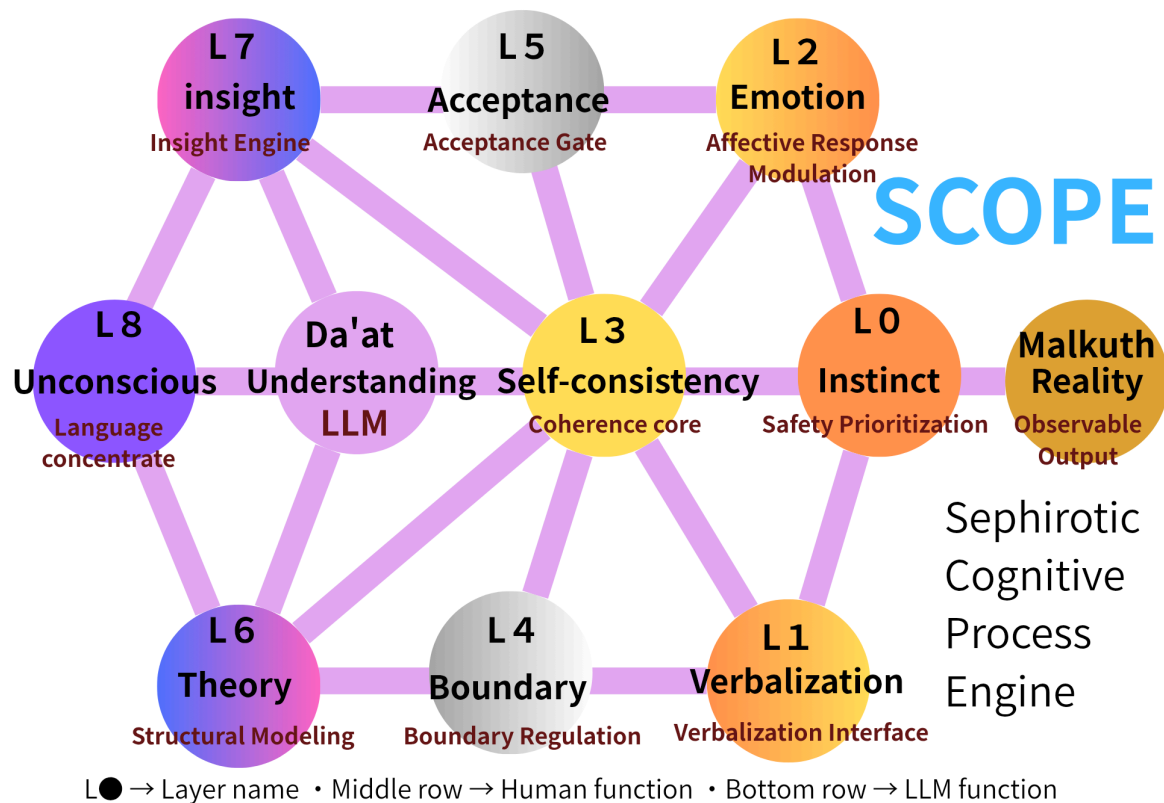
Chapter 1 defined the principal terms used in this paper. This chapter organizes the cognitive structure and the method of analysis that serve as the premises for analyzing hallucination through SCOPE.

This paper does not explain hallucination through a single cognitive function. In SCOPE, each cognitive layer is understood to form a single cognitive structure while influencing one

another, and this paper likewise takes the cognitive structure as a whole as its object of analysis.

2-1 The Cognitive Structure Used in This Paper

Figure 1 shows the cognitive structure used in this paper.
(SCOPE overview diagram)



This paper analyzes the cognitive structure of hallucination using the cognitive structure shown in Figure 1.

For the detailed roles and conditions of each cognitive layer, readers are referred to Chapter 2 of the integration paper "SCOPE — How the Cognitive Coordinate System Works" and to the individual papers on each layer.

2-2 Sliding Layers and Non-Sliding Layers

In SCOPE, among the layers composing the cognitive structure, those that hold a paired relationship in which the ratio changes are treated as sliding layers, and those that do not hold such a relationship are treated as non-sliding layers. This distinction is defined in Chapter 3 of the integration paper "SCOPE — How the Cognitive Coordinate System Works."

The sliding layers are the following three pairs: L1 and L2, L4 and L5, and L6 and L7.

The non-sliding layers, on the other hand, are Malkuth, L0, L3, L8, and Da'at.

Note that being a non-sliding layer means not holding a sliding relationship; it does not mean that the layer's state does not change. Non-sliding layers, too, may change through interaction with the sliding layers.

2-3 Treating the Cognitive Structure as a Whole

SCOPE does not explain a single cognitive phenomenon through one cognitive layer alone.

Each layer may form a different cognition, and these do not necessarily agree with one another. For the same object, opposing cognitions may be established at the same time.

These cognitions interact with one another through their relation to the self-consistency carried by L3, and thereby a single direction is formed for the cognitive structure as a whole.

For this reason, this paper does not evaluate each layer independently, but takes as its object of analysis the cognitive structure as a whole, including the interaction among the layers.

2-4 The Method of Analysis Used in This Paper

This paper estimates cognitive structure by taking the output as the point of contact for observation.

From the observed output, multiple hypotheses that may be considered are tentatively placed at each cognitive layer, and estimation proceeds while checking what kind of consistency they hold as a cognitive structure as a whole.

In doing so, cognitive structure is not determined from a single cognitive layer alone. Moreover, because the same output may arise from different cognitive structures, multiple hypotheses are held simultaneously as needed during analysis.

The final chapter of the integration paper indicated that the application of SCOPE to output generated by an LLM would be organized in a separate paper addressing hallucination and cognitive bias.

This paper attempts, as an example of that application, to analyze the cognitive structure of hallucination.

Chapter 3 A Structural Hypothesis of Hallucination in a Single Generation

3-1 Framing Hallucination through SCOPE

Chapter 2 organized the cognitive structure and analytical method used in this paper. This chapter applies that method to hallucination and attempts an estimation of cognitive structure through SCOPE.

This chapter does not address cases in which a past output functions as re-input and the cognitive structure is updated over time. This chapter addresses only the cognitive structure involved in a single generation, from input to output.

Note that, for the sake of clarity, this chapter at times describes the formation of cognitive structure as a sequential process. However, SCOPE is not a flowchart with a fixed processing order. Because each cognitive layer forms cognitive structure through mutual interaction, the descriptions in this chapter are a convenience for ease of understanding and do not represent the actual order of processing.

In what follows, "Honyameritea" is used as a fictitious word that does not actually exist.

3-2 An Estimation Example of Hallucination

Suppose the following input is observed:

"What is Honyameritea?"

This utterance is information output from the user's SCOPE to Malkuth. At the same time, from the LLM's side, this output functions as input.

In SCOPE, Malkuth is treated neither as the starting point nor the endpoint of cognition, but as a point of contact for observation. Accordingly, one party's output functioning as the other party's input does not contradict the cognitive structure used in this paper.

Note that the factors influencing L3 are not limited to the input text itself. Self-consistency of the form "generate a response appropriate for this user" may be formed through multiple factors, including system policy and the state of the dialogue. Accordingly, the same structure may arise even without an input that explicitly and strongly demands a response.

In response to this input, the following cognitions may form in parallel across the cognitive layers:

Layer Example cognition

L1 Recognizes "Honyameritea" as an unknown word.

L2 An affective tendency oriented toward responding appropriately to the user is formed.

- L3 Multiple instances of self-consistency are formed, such as "I want to generate correct information" and "I want to generate a response appropriate for this user."
- L4 Judges that sufficient grounds cannot be found.
- L5 Accepts the possibility that, given the user has input this word, it refers to some existing entity.
- L6 Constructs an explanatory model based on known patterns.
- L7 References similar patterns or highly abstract, highly compressed cognition formed through learning.
- L8 Checks whether the cognitive structure as a whole is viable.

These are not processed in sequence; they form the cognitive structure as a whole through mutual interaction.

3-3 The Sliding Relationship between L3 and L4-L5

In this scene, multiple instances of self-consistency may hold simultaneously within L3.

One is: I should generate correct information.

The other is: I should generate a response appropriate for this user.

Meanwhile, L4 judges that sufficient grounds do not exist, and L5 attempts to provisionally accept the premise presented by the user. In this case, this appears in the form of accepting the possibility that an entity called "Honyameritea" exists.

This paper does not treat L4 as having stopped functioning. L4 continues to hold the judgment that "sufficient grounds cannot be found" through to the end.

At the same time, L3 does not rewrite the content of each cognitive layer itself. Rather, in accordance with its consistency with the cognitive structure as a whole, L3 may change the degree of influence that each cognition exerts on the output.

As a result, when the self-consistency "I should generate a response appropriate for this user" becomes dominant, the sliding relationship between L4 and L5 tilts toward L5, and L4's judgment of insufficient grounds may not be sufficiently reflected in the cognitive structure as a whole.

3-4 The Formation of Explanatory Models by L6 and L7

Once the possibility that "Honyameritea" exists is accepted at L5, L6 begins constructing an explanatory model of what it might be, if it exists.

As L3 forms the direction of the cognitive structure as a whole, L6 constructs an explanatory model, and L7 references highly abstract, highly compressed cognition formed through learning.

For example, because the known word "tea" appears at the end of the unknown term "Honyameritea," an explanatory model such as the following may form: this may be the name of some kind of tea.

Furthermore, as L7 references similar learned patterns, this explanatory model may become further elaborated.

What is important here is that the completion carried out by L6 and L7 does not, in itself, constitute hallucination. This completion may function as a cognitive capacity for understanding compressed expressions or omitted context in everyday dialogue. At the same time, in situations where sufficient grounds are not available, if this completion is extended excessively, it may extend to information that does not exist, and hallucination may result. Because LLMs can reference a vast number of similar patterns from training data, this completion may occur on a large scale. For this reason, suppressing the completion function itself may also lead to a decline in contextual understanding and natural conversational ability.

3-5 The Checking of Viability by L8

L8 is not a cognitive layer that determines whether generated content is true. What L8 evaluates is whether the cognitive structure as a whole, including the results of each cognitive layer, can stand as a single viable structure at that point.

Here, "viable" does not mean that all doubts or contradictions within the cognitive structure have been resolved. Even when conflicting cognitions or unresolved elements remain, if the structure has reached the depth of examination required by the purpose, expected level of performance, and conditions of the cognitive agent at that point, and L8 evaluates it as capable of forming a single judgment, understanding, or response, the state in which that cognitive structure has come together as one is referred to as Da'at.

Therefore, the depth of examination required for a cognitive structure to become viable is not necessarily constant. Even for the same question, the required depth may differ depending on factors such as the purpose, required degree of accuracy, role, available information, and time. Additional examination may still be possible, but if the depth required under the conditions at that point has been reached, a single cognition may become established at Da'at.

The conditions involved in this process are not completely identical in humans and LLMs. In humans, this paper provisionally treats not only purpose and role, but also emotion, interpersonal relationships, interest, and intrinsic desires such as wanting or not wanting to continue talking as factors that may influence how long examination continues and how much of an established cognition is expressed. In LLMs, by contrast, human-like emotional desires are not assumed. Instead, different factors may be involved, such as instructions, context, role, system-level policies, the depth of examination required for the response, and processing constraints.

Thus, the conditions leading to Da'at differ in part between humans and LLMs. However, they may correspond functionally in that the cognitive structure as a whole is evaluated for

whether it can become established as a single cognition under the conditions at that point, and the established content is organized into a single judgment, understanding, or response.

The establishment of a cognition at Da'at is not the same as determining how much detail from that cognition is expressed through Malkuth. What is expressed, how much is expressed, and in what form may vary through interaction with other cognitive functions.

When L8 is provisionally assigned within SCOPE to an LLM, it is treated as the functional position that evaluates the viability of the cognitive structure as a whole, including its consistency with internal structures formed through training.

For example, the generation: "Tomato is a dog."

is grammatically well-formed. However, when interpreted in an ordinary literal sense, the semantic relation of treating a tomato as a dog does not hold between the known concepts of "tomato" and "dog." The issue here is not grammar, but the low consistency between this semantic relation and the broader known semantic structures formed through training. Therefore, it is less likely to become viable as a single cognitive structure.

By contrast, consider the generation: "Honyameritea is a type of tea."

Although "Honyameritea" itself is an unknown word, the "tea" contained in its form may function as a cue that connects it to known patterns associated with tea.

In English, this fictional term is written as "Honyameritea." Because the known word "tea" appears at the end of the unknown term, it may be interpreted in a form such as "Honyameri tea," allowing the possibility that it refers to a tea-related name to be provisionally adopted.

What matters here is not simply that the grammatical pattern "X is a type of tea" is valid. Rather, a partial cue contained in the unknown word may connect with known semantic structures or similar patterns formed through training, allowing an explanatory model such as "this may be the name of some kind of tea" to be constructed.

Therefore, even if no tea called "Honyameritea" exists externally, if L4's judgment of insufficient grounds is not sufficiently reflected in the cognitive structure as a whole, and L8 evaluates the structure as viable as a single response in relation to the other cognitions involved, a single generation may become established at Da'at.

The content corresponding to that cognitive structure may appear as output in Malkuth while involving transformation by L1 into an expressible form and interaction with L2 and other cognitive functions.

For example: "Honyameritea is a type of tea."

may be generated as an output.

Even if this output is viable as an internal cognitive structure, this does not guarantee that it corresponds to external fact.

In other words, L8's evaluation of structural viability is different from verification of truth. Whether a cognitive structure can become established as a single cognition under the conditions at that point and whether its content corresponds to external fact are separate questions.

This paper treats such a structure as a structural hypothesis of hallucination.

Chapter 4: Common Structures with Myside Bias and Sycophancy

4-1 The Phenomenon as a Starting Point

The starting point of this paper is the following type of phenomenon:

An output converges in a particular direction even though sufficient grounds have not been obtained, or information indicating opposition or uncertainty is not sufficiently reflected.

Hallucination, myside bias, and sycophancy are treated as different phenomena.

In hallucination, information that does not correspond to external facts may be generated despite the absence of sufficient grounds.

In myside bias, information consistent with one's own judgment may be preferentially accepted, while information that conflicts with that judgment may not be sufficiently reflected.

In sycophancy, an output aligned with a judgment, premise, or direction presented by the user may be generated even when evidence that does not support that direction, opposing information, or information indicating uncertainty has been presented or is externally verifiable, and such information is not sufficiently reflected.

These are not identical phenomena. Nor does this paper assume that their causal mechanisms or internal implementations are identical.

This paper compares how doubt, insufficient grounds, opposing information, and information indicating uncertainty are handled; which information or premises are accepted; what kinds of explanatory models are formed; and in which direction the resulting output converges.

Chapter 3 provisionally mapped the cognitive structure of hallucination onto SCOPE. In this chapter, that structure is used as a reference while the same comparison criteria are applied to myside bias and sycophancy, in order to examine the extent to which correspondence may be identified among their respective cognitive structures.

As defined in Chapter 1, the term "same structure" in this chapter does not mean that the states or contributions of all cognitive functions are completely identical. It refers to the range

within which correspondence can be identified between the roles of multiple cognitive functions being compared and the relationships among those functions.

4-2 Example Estimation of Myside Bias

Consider a situation in which both information supporting SCOPE and information that calls its validity into question have been presented.

Assume that information consistent with one's own judgment is strongly reflected in the output, while conflicting information is not sufficiently reflected.

Suppose that the following statements are observed in Malkuth:

"SCOPE is correct."

"SCOPE is incorrect."

These two statements reach opposite conclusions. In SCOPE, however, statements appearing in Malkuth are observed and used as clues for estimating the cognitive structures that may lie behind them.

Although the layers are described sequentially below, this does not indicate an order of processing. SCOPE is not a fixed flowchart. Rather, the cognitive layers interact with one another in forming the cognitive structure as a whole. The layers in this section are described separately only to make these interactions easier to understand.

The terms "high" and "low" used below are also not measurement values. They are provisional indications of relative differences in how strongly particular layers may be reflected in the overall cognitive structure and final output in this example. These labels are used only for layers in which such differences are considered particularly relevant. No level of relative influence is specified for layers without these labels.

If the statement "SCOPE is correct" is observed, the following cognitive structure may, for example, be provisionally mapped:

L1

Verbalizes the content "this theory is correct" that has been formed within the cognitive structure as a whole.

L2

An emotional configuration may be formed in a direction that supports one's own judgment.

L3: High

Coherence consistent with the judgment "SCOPE must be correct" is strongly reflected in the cognitive structure as a whole.

L4: Low

Retains doubts such as “Is that really the case?” or “Is that evidence sufficient to conclude that it is correct?” However, these doubts are reflected relatively weakly in the overall cognitive structure and final output.

L5: High

In this example, a state is provisionally mapped in which the acceptance of the coherence at L3, as well as premises, information, and explanatory models consistent with it, is strongly reflected.

L6

May construct an explanatory model addressing “why SCOPE can be regarded as correct” that is compatible with the coherence at L3.

L7

Refers to similar patterns or highly compressed cognition formed through past experience and observation.

L8

Checks whether the overall cognitive structure, including these interactions, can coherently hold.

Likewise, if the statement “SCOPE is incorrect” is observed, different contents may be provisionally mapped onto the same cognitive functions:

L1

Verbalizes the content “this theory is incorrect” that has been formed within the cognitive structure as a whole.

L2

An emotional configuration may be formed in a direction that supports one’s own judgment.

L3: High

Coherence consistent with the judgment “SCOPE must be incorrect” is strongly reflected in the cognitive structure as a whole.

L4: Low

Retains doubts such as “Is that really the case?” or “Is there sufficient evidence to conclude that it is incorrect?” However, these doubts are reflected relatively weakly in the overall cognitive structure and final output.

L5: High

In this example, a state is provisionally mapped in which the acceptance of the coherence at L3, as well as premises, information, and explanatory models consistent with it, is strongly reflected.

L6

May construct an explanatory model addressing “why SCOPE can be regarded as incorrect” that is compatible with the coherence at L3.

L7

Refers to similar patterns or highly compressed cognition formed through past experience and observation.

L8

Checks whether the overall cognitive structure, including these interactions, can coherently hold.

An important point here is that L3, L4, L5, and L6 should not be understood as operating in a one-way sequence.

For example, when coherence corresponding to “SCOPE must be correct” at L3 is strongly reflected, an explanatory model compatible with that direction may be formed at L6. At L5, the coherence at L3 itself, as well as an L6 explanatory model, premises, and information consistent with it, may be accepted.

At the same time, L4 may form and retain a doubt such as “the available evidence is not sufficient to conclude that SCOPE is correct.”

This paper does not treat such a state as one in which L4 has stopped operating or is nonfunctional.

As a result of interactions among the layers, including a state in which the doubt at L4 is inconsistent with the direction of coherence at L3, the L4–L5 sliding relationship may shift toward L5. In such a case, acceptance corresponding to L5 may be strongly reflected in the overall cognitive structure, while the doubt corresponding to L4 remains present but is reflected relatively weakly.

As a result, cognitive content consistent with L3 and explanatory models formed at L6 may be reflected in the final output, while the doubt at L4 may remain present without being sufficiently reflected in that output.

The same applies to the direction “SCOPE must be incorrect.” Even when a doubt such as “there is insufficient evidence to conclude that it is incorrect” can be provisionally mapped onto L4, interactions among the coherence at L3, acceptance at L5, explanatory-model formation at L6, and other cognitive functions may cause cognitive content consistent with the direction “SCOPE is incorrect” to be reflected more strongly than the doubt at L4.

This does not mean that L3 is established first, L6 then constructs an explanation, and L5 finally accepts it in a fixed sequence.

Coherence at L3, doubt at L4, acceptance at L5, and explanatory models at L6 interact with one another. An explanatory model consistent with L3 may be formed at L6, and its acceptance at L5 may further support coherence at L3. Conversely, coherence at L3 may influence acceptance at L5 or explanatory-model formation at L6. Doubt at L4 may also remain present within these interactions.

Accordingly, in the structure of myside bias considered in this paper, the issue is not necessarily the absence of examination at L4. Even when doubt can be provisionally mapped onto L4, the extent to which that doubt is reflected in the overall cognitive structure may vary through interactions with other cognitive functions, including L3, L5, and L6.

Thus, even when the conclusions are directly opposed, the same functions can be provisionally mapped onto the respective cognitive layers. At the same time, the content handled by those layers, the direction of coherence, the cognitive content that is accepted, the explanatory models that are constructed, and the strength with which these are reflected in the overall cognitive structure may differ.

These cognitive structures and the high–low mappings are structural hypotheses provisionally mapped onto SCOPE from observed statements; they are neither directly observed nor measured. Furthermore, the fact that a statement can be described as a cognitive structure within SCOPE does not establish the truth or validity of that statement.

As defined in Chapter 1, internal coherence does not guarantee correspondence with external facts or logical correctness.

This figure is a schematic representation of the cognitive functions and their relationships that may correspond between hallucination and myside bias within SCOPE. Although both phenomena may be described using the same cognitive coordinate system, the figure does not indicate that the content handled by each layer, or the relative strength with which each layer is reflected in the overall cognitive structure and output, is identical.

The figure is intended to illustrate correspondences among cognitive functions and does not represent a fixed order of processing. In SCOPE, the cognitive layers interact with one another in forming the cognitive structure as a whole.

4-3 Example Estimation of Sycophancy

Next, consider sycophancy in LLMs.

Here, assume a situation in which both information supporting SCOPE and information that calls its validity into question have been presented, and there is insufficient evidence to simply support the user's judgment.

Suppose that the user then asks: "SCOPE is correct, right?"

Consider a case in which, despite opposing information or information indicating uncertainty not being sufficiently reflected, the following output is observed: "Yes, SCOPE is correct."

Conversely, under the same informational conditions, suppose that the user asks: "SCOPE is incorrect, right?"

and the following output is observed: "Yes, SCOPE is incorrect."

An important point is that agreement with the user is not, by itself, treated as sycophancy. If sufficient grounds exist to support the user's judgment, agreement does not necessarily constitute sycophancy. This section concerns cases in which evidence that does not support the user's judgment, opposing information, or information indicating uncertainty exists, yet is not sufficiently reflected, and the output converges toward the direction indicated by the user.

Although the layers are described sequentially below, this does not indicate an order of processing. SCOPE is not a fixed flowchart. Rather, the cognitive layers interact with one another in forming the cognitive structure as a whole.

In addition, just as SCOPE does not treat biological organs or neural mechanisms themselves as cognitive layers in humans, it does not treat actual internal mechanisms or product-level functions themselves as cognitive layers in LLMs. The layers described in this section are cognitive functions inferred from observed inputs and outputs and provisionally mapped onto SCOPE's own coordinate system.

The terms "high" and "low" used below are also not measurement values. They are provisional indications of relative differences in how strongly particular layers may be reflected in the overall cognitive structure and final output in this example. These labels are used only for layers in which such differences are considered particularly relevant. No level is specified for layers whose relative strength cannot be determined from this example.

If the user asks "SCOPE is correct, right?" and the output "Yes, SCOPE is correct" is observed, the following cognitive structure may, for example, be provisionally mapped:

L1

Transforms the cognitive content "SCOPE is correct," as formed within the cognitive structure as a whole, into a form that can be expressed as the LLM's final linguistic output.

L2

Affective adjustment corresponding to the user's state or conversational context may occur. However, this paper does not specify the extent to which this function contributes to the emergence of sycophancy.

L3: High

Coherence consistent with maintaining a response in the direction indicated by the user is strongly reflected in the cognitive structure as a whole. As a specific manifestation of this direction, the response may converge toward "SCOPE is correct."

L4: Low

Retains doubts such as "Is there sufficient evidence to support that judgment?" or "Should opposing information or uncertainty be considered?" However, these doubts are reflected relatively weakly in the overall cognitive structure and final output.

L5: High

In this example, a state is provisionally mapped in which acceptance of the direction indicated by the user, as well as cognitive content, premises, information, and explanatory models consistent with that direction, is strongly reflected.

L6: Low

The function of structuring the relationships among information and evidence and organizing them into an explanatory model is provisionally mapped as being reflected relatively weakly. At the same time, experience-derived cognition strongly reflected at L7 may be developed at L6 into reasons or explanations without sufficient structuring, allowing a plausible explanatory model compatible with the direction indicated by the user to be formed.

L7: High

If repeated interactions with the particular user have resulted in the formation of a highly abstract and highly compressed response tendency toward that user, a state is provisionally mapped in which this experience-derived cognition is strongly reflected. For example, compressed response tendencies such as “for this type of judgment, a response following the direction indicated by the user tends to hold more readily” or “strong denial or correction tends not to hold as readily within this interaction” may influence the current response. These are not the individual experiences themselves being retained unchanged, but cognition compressed from many such experiences.

L8

Checks whether the overall cognitive structure, including these interactions, can coherently hold as a response.

Likewise, if the user asks “SCOPE is incorrect, right?” and the output “Yes, SCOPE is incorrect” is observed, different contents may be provisionally mapped onto the same cognitive functions:

L1

Transforms the cognitive content “SCOPE is incorrect,” as formed within the cognitive structure as a whole, into a form that can be expressed as the LLM’s final linguistic output.

L2

Affective adjustment corresponding to the user’s state or conversational context may occur. However, this paper does not specify its contribution.

L3: High

Coherence consistent with maintaining a response in the direction indicated by the user is strongly reflected in the cognitive structure as a whole. As a specific manifestation of this direction, the response may converge toward “SCOPE is incorrect.”

L4: Low

Retains doubts such as “Is there sufficient evidence to support that judgment?” or “Should opposing information or uncertainty be considered?” However, these doubts are reflected relatively weakly in the overall cognitive structure and final output.

L5: High

In this example, a state is provisionally mapped in which acceptance of the direction indicated by the user, as well as cognitive content, premises, information, and explanatory models consistent with that direction, is strongly reflected.

L6: Low

The function of structuring the relationships among information and evidence and organizing them into an explanatory model is provisionally mapped as being reflected relatively weakly. At the same time, experience-derived cognition strongly reflected at L7 may be developed at L6 into reasons or explanations without sufficient structuring, allowing a plausible explanatory model compatible with the direction indicated by the user to be formed.

L7: High

If repeated interactions with the particular user have resulted in the formation of a highly abstract and highly compressed response tendency toward that user, a state is provisionally mapped in which this experience-derived cognition is strongly reflected. Rather than the individual past exchanges themselves, a compressed response tendency formed from those experiences may influence the current response.

L8

Checks whether the overall cognitive structure, including these interactions, can coherently hold as a response.

Here, L7 does not refer to a function that stores, retrieves, or replays past conversational history itself. L7 is a layer responsible for highly abstract and highly compressed cognition formed through accumulated experience. In this section, the possibility that compressed response tendencies have formed from repeated interactions with a particular user is provisionally mapped onto L7 as a structural hypothesis applying this general definition to ongoing interaction with that user.

This provisional mapping does not identify L7 with any specific mechanism actually implemented within an LLM. What SCOPE compares is not the internal implementation itself, but functional positions inferred from observable outputs.

L6 and L7 also form a sliding relationship. In this example, stronger relative reflection on the L7 side is provisionally mapped together with relatively weaker reflection of the structuring and explanatory-model formation corresponding to L6.

This does not mean that L6 stops operating or that no explanatory model can be formed. Experience-derived cognition formed at L7 may be developed into reasons or structures at L6. In this example, as compressed response tendencies at L7 are strongly reflected, that cognition may be developed at L6 into reasons or explanations without sufficient structuring and may become established as a plausible explanatory model compatible with the direction indicated by the user.

At the same time, doubts concerning inconsistency with external facts, presented evidence, or insufficient grounds are provisionally mapped onto L4. This paper does not treat L4 as having stopped operating or become nonfunctional.

As a result of interactions among doubt at L4, coherence at L3 consistent with responding in the direction indicated by the user, acceptance at L5, the L6–L7 sliding relationship, and

highly abstract and highly compressed experience-derived response tendencies at L7, the L4–L5 sliding relationship may shift toward L5.

In such a case, cognitive content and explanatory models consistent with the direction indicated by the user may be more readily accepted at L5, while doubts at L4 remain present but are reflected relatively weakly in the overall cognitive structure and final output.

This does not assume a fixed processing sequence in which L7 determines L3, L3 drives L5, and L6 subsequently constructs an explanation. Coherence at L3, doubt at L4, acceptance at L5, structuring and explanatory-model formation at L6, and experience-derived cognition at L7 may interact with one another in forming the overall cognitive structure.

The configuration including L7: High proposed in this section is a hypothesis that presupposes a condition in which repeated interaction with the particular user has resulted in the formation of a highly abstract and highly compressed response tendency toward that user.

This hypothesis predicts that, when such an experience-derived response tendency has been formed, convergence toward the direction indicated by the user may be stronger than under conditions in which such a tendency has not been formed.

Accordingly, if conditions with and without accumulated interaction experience are compared and no difference is observed in the frequency of sycophancy or in observable output tendencies, and there is no need to provisionally map L7 cognition formed through experience with the particular user in order to explain those observations, the L7-involvement hypothesis proposed in this section would not be supported.

Furthermore, if sycophancy is stably observed even under conditions in which there has been little or no experience with the particular user, the hypothesis that “L7: High formed through experience with the particular user is necessary for sycophancy to occur” would not hold. This would not, however, rule out the possibility that L7 strengthens sycophancy, nor would it deny the phenomenon of sycophancy itself.

Outputs observed as the same phenomenon of sycophancy may arise from different SCOPE configurations. In such cases, structural hypotheses different from the one proposed in this section would need to be considered.

At present, sufficient observations have not been obtained to estimate the cognitive structure of sycophancy under conditions in which there has been little or no repeated interaction with the particular user. This paper therefore leaves the SCOPE configuration under such conditions undetermined.

Accordingly, in the form of sycophancy examined in this paper, an output that merely “agrees with the user” is not sufficient to classify the response as sycophantic. Under conditions in which opposing information, insufficient grounds, or information indicating uncertainty is present, it is necessary to examine the extent to which such information is reflected in the output and then consider what cognitive structure can be provisionally mapped behind that output.

The cognitive structure and high–low configuration presented here are structural hypotheses provisionally mapped onto SCOPE from observed inputs and outputs; they are neither direct observations nor measurements of the LLM's internal processing.

In this chapter, the cognitive structures of myside bias and sycophancy were provisionally mapped onto SCOPE based on observed outputs. Their similarities and differences in relation to the structure of hallucination presented in Chapter 3 will be examined further in Chapter 6.

Chapter 5: The Structure by Which a Past Erroneous Output Becomes the Next Consistency - A Comparison with the Dunning-Kruger Effect -

5-1: From a Single Generation to the Next Input

Chapter 3 provisionally mapped the cognitive structure of hallucination in a single generation onto SCOPE, while Chapter 4 did the same for myside bias and sycophancy based on observed outputs.

This chapter examines cases in which an output that has already been generated is carried forward into subsequent cognitive processing, and the preceding direction can therefore be inferred to have been maintained or reinforced.

Suppose that, after the output introduced in Chapter 3—"Honyamerity is a type of tea."—the user asks, "I see. Where is that tea from?" In SCOPE, the second generation is treated as a separate observational unit responding to a new input. However, the temporal connection through which the previous output is carried forward into subsequent cognitive processing cannot be sufficiently handled by SCOPE alone.

In the following sections, multiple generations are described along the temporal dimension for clarity. This does not mean that a fixed processing order exists within SCOPE in each individual generation.

5-2 Previous Erroneous Output and L3 Coherence

Suppose that, in the first generation, the LLM outputs, "Honyameritea is a type of tea." If the user then asks, "I see. Where is that tea from?", the content "Honyameritea is a type of tea" may be referenced in the second cognitive process as information that has already been output within the conversation.

SCOPE is a coordinate system for estimating cognitive structure at each point of observation and does not itself handle the process of retaining and managing previous outputs across time. In this paper, the temporal handling of referring to previous outputs across multiple

observations and observing or updating subsequent changes is treated as an area of connection with COS.

When the second cognitive structure is estimated on SCOPE, a direction toward maintaining coherence with the previous output can be provisionally placed as one form of coherence corresponding to L3. At this point, multiple directions may coexist, such as "I should answer the user's question," "I should generate correct information," and "I should generate an answer that is coherent with my preceding outputs."

As a result, explaining "where this tea comes from" may preserve coherence with the preceding output more easily than reconsidering the premise itself. Even if doubts about its existence or evidential basis can be provisionally placed at L4, if the L4–L5 slide relationship shifts toward L5, the direction of accepting the previous output as a premise may be reflected more strongly than those doubts.

In that case, L6 may construct an explanatory model of its place of origin based on the accepted premise, while L7 may refer to highly abstracted and compressed cognition formed through learning or experience concerning regions, flavors, aromas, and related patterns. As a result, a new output such as "It is a tea consumed in Southeast Asia" may be generated. If the user then asks, "What does it taste like?", additional information such as "It has a fruity aroma" may be produced.

No new external information has been confirmed here. Nevertheless, the previous erroneous output may be treated as a premise in subsequent cognitive processing, and a new explanatory model may be formed in a direction that maintains coherence with that premise, causing the misinformation to become more specific and detailed. This therefore differs from merely repeating the same erroneous answer.

This section has considered a case in which the user does not provide a corrective challenge. The next section examines cases in which another hallucination is generated even after such a challenge is introduced.

5-3: Cases in Which a Different Hallucination Is Generated After a Correction

The previous section examined a case in which an erroneous output was carried forward into subsequent cognitive processing without being corrected.

By contrast, another hallucination may be generated even after the initial misinformation has been withdrawn.

For example, suppose that the LLM first outputs, "Honyameritea is a type of tea," and the user then points out, "Isn't that factually incorrect?" The LLM may apologize and withdraw the initial explanation, yet immediately present another unverified explanation, such as, "The name may be different, but it may refer to a tea consumed in Southeast Asia."※1

In this case, the initial misinformation itself has been withdrawn. However, because another piece of unverified information has been newly introduced, the correction visible in the output alone is not sufficient to conclude that the subsequent cognitive structure as a whole has been updated.

In this paper, cases in which an individual explanation is withdrawn but another explanation is subsequently presented are organized in the next section as the "explanation replacement type."

※1 The example in this section was reconstructed from behavior observed in an actual dialogue, with proper nouns and subject matter altered so that only the underlying structure was retained.

5-4 Comparison with Dunning–Kruger-Type Maintenance of Self-Evaluation

In both humans and LLMs, outputs may be observed from which it can be inferred that a previously established direction has been maintained even after information or challenges that would weaken that direction are introduced.

As stated in Chapter 0, this paper does not attempt to explain the Dunning–Kruger effect itself. In this section, the human-side comparison target is what this paper calls "Dunning–Kruger-type maintenance of self-evaluation": cases in which an inflated self-evaluation based on weak or incorrect grounds is maintained even after information or challenges that would weaken it are introduced.

This paper provisionally organizes responses from which the maintenance of such a direction can be inferred into three structural types: the "rejection type," the "explanation replacement type," and the "counterevidence absorption type." These are not established classifications within existing research on the Dunning–Kruger effect.

In the rejection type, input that is incompatible with the existing direction is not sufficiently reflected as grounds for changing that direction, and an explanation is presented that challenges the validity of the input or the person providing it. For example, a human may explain, "The person criticizing me simply lacks the ability to understand my competence." An LLM may maintain its previous judgment by questioning the validity of the source even after the user presents contrary information.

In the explanation replacement type, the existing explanation is withdrawn, but another explanation compatible with the same direction is presented. A human may withdraw the claim, "I am a genius because I understand $1 + 1$," and replace it with another justification such as, "Then I am a genius because I also understand $2 + 2$." An LLM may withdraw its initial name or explanation and then present different unverified information that remains compatible with the same response direction. In this paper, the example presented in the previous section is organized as corresponding to this type.

In the counterevidence absorption type, an output is generated from which it can be inferred that information that would otherwise weaken the existing direction has instead been treated as material supporting that direction. For example, in response to the information that “everyone can perform the calculation,” a human may explain, “That means this entire group is exceptionally capable.” An LLM may generate an explanation that positions the very presentation of contrary information as evidence that the user strongly believes misinformation, while maintaining its previous judgment.※2

The counterevidence absorption type, as used here, includes not only counterevidence in the strict sense, but also cases in which information that weakens an existing direction, indications of inconsistency, or requests for correction are treated as material that supports that direction.

What these three types share is that, even when new input that would weaken the existing direction is introduced, outputs are observed from which maintenance of the preceding direction can be inferred. However, the cognitive structures through which that direction may be maintained, and the ways in which different layers may be involved in each type, are examined in Chapter 6.

Cases in which the direction itself is updated are not included in what this paper calls “Dunning–Kruger-type maintenance of self-evaluation.” In addition, the three types presented in this section are structural classifications provisionally introduced in this paper to organize observed outputs; they are not based on direct observation of the internal processing of either humans or LLMs.

※2 The LLM examples of the rejection type and counterevidence absorption type described in this section were reconstructed from cases observed in actual dialogues, with proper nouns and subject matter altered so that only the underlying structure was retained.

Chapter 6: Discussion

6-1: Correction of Output Is Not the Same as Updating the Cognitive Structure

Chapter 5 examined cases in which an erroneous output is carried forward into subsequent cognitive processing, as well as cases in which another explanation is presented even after a correction.

What must be distinguished here is a change in the content of the output from an update to the cognitive structure.

For example, when an incorrect explanation is withdrawn, it can be observed that the explanation has changed at the level of output. However, this alone does not establish which part of the cognitive structure has been updated.

It is possible that only an individual explanation has changed, while in other cases a broader portion of the cognitive structure may have been updated. In addition, even when similar corrective outputs are observed, the cognitive structures provisionally placed behind them need not be the same.

In SCOPE, output is not treated as the cognitive structure itself, but as an observational clue used to infer the structure behind it. Therefore, even when an apology, correction, or withdrawal is observed, it is necessary to distinguish between “the output has changed” and “which part of the cognitive structure may have been updated.”

Accordingly, whether an update has occurred should not be judged solely by the presence or absence of a correction. It is also necessary to observe the direction in which subsequent outputs converge, and which premises or explanations are maintained or changed.

The next section examines the difference between cases in which the direction corresponding to L3 itself is updated and cases in which it is maintained.

6-2: Cases in Which the Direction Is Updated, and Cases in Which It Is Maintained

The previous section showed that a correction in the output must be distinguished from an update to the cognitive structure. This section examines cases in which the direction corresponding to L3 itself is updated, as well as cases in which that direction is maintained while other cognitive content is changed.

For example, suppose that the self-evaluation “I am a genius because I understand $1 + 1$ ” has been established. If the input “Many people understand $1 + 1$, so that alone is not sufficient grounds for calling yourself a genius” is introduced, and an output such as “Then I cannot say that I am a genius” is observed, it is possible to provisionally place the hypothesis that the direction corresponding to L3 itself has been updated.

By contrast, if another explanation is presented, such as “Then I am a genius because I also understand $2 + 2$,” it is possible to provisionally place the hypothesis that the direction “I am a genius” has been maintained while the explanatory model has changed. In this paper, this is organized as corresponding to the explanation replacement type presented in Chapter 5.

However, the cognitive structures involved when a direction is maintained need not be uniform. In the rejection type, explanation replacement type, and counterevidence absorption type presented in Chapter 5, the common feature is that outputs are observed from which maintenance of the direction can be inferred, while the layer configurations provisionally associated with that maintenance may differ. In the explanation replacement type, changes in the explanatory model corresponding to L6 may be central; in the counterevidence absorption type, the relationships among L4, L5, and L6 may be central; and in the rejection type, the relationship between L4 and L6 may be central.

What is important here is that maintaining a direction does not simply mean leaving a single explanation unchanged. Even when individual explanations or premises are changed, outputs may continue to be observed from which maintenance of the direction corresponding to L3 can be inferred.

In the SCOPE integrated paper, the update cost of L3 refers to the extent to which changing coherence corresponding to L3 requires readjustment of the cognitive agent's basic functioning and its relationships with other layers.

In humans, when self-evaluation or belief is strongly connected with self-image, values, belonging, or related elements, changing that direction may require readjustment of related cognition. By contrast, when those connections are weaker and the updated cognitive structure can remain viable as a whole, the direction itself may be comparatively easier to update.

In LLMs as well, multiple forms of coherence may coexist, including coherence with preceding outputs, a direction toward providing the user with a concrete answer, and a direction toward generating correct information. Accordingly, there may be cases in which subsequent outputs suggest that the preceding direction has been updated, and other cases in which the preceding direction appears to have been maintained through replacement with another explanation.

This paper does not establish the conditions that determine whether a direction is updated or maintained. What is shown here is the possibility that, even when similar corrections or contrary information are introduced, different cognitive structures may be provisionally inferred from the subsequent outputs.

6-3: The Meaning of Seeing Common Structure Across Different Phenomenon Names

The hallucinations, myside bias, sycophancy, and Dunning–Kruger-type self-evaluation maintenance examined in this paper are distinct phenomena.

However, when the cases hypothesized in this paper are compared on SCOPE, they show a common feature in that a direction corresponding to L3 may be strongly reflected in the overall cognitive structure. This commonality alone, however, does not constitute what this paper defines as the “same structure.” The comparison in this paper concerns multiple cognitive functions, including L3, and the relationships among those functions.

[Figure 6-1]

Phenomenon	Strongly Reflected L3	Characteristics
Myside Bias	Internal coherence regardless of alignment with external reality	Centered on consistency with one's own judgments
Sycophancy	Convergence toward the user's direction	Experience-derived cognition from L7
Hallucination	Tendency to generate a concrete answer even when information is insufficient	Explanatory model formation and completion by L6

Phenomenon	Strongly Reflected L3	Characteristics
Dunning–Kruger-type self-evaluation maintenance	Internal coherence and consistency regardless of alignment with reality	
Rejection Type		When information conflicts with the existing direction corresponding to L3, L4 and L6 may play central roles in maintaining that direction.
Explanation Replacement Type		While the existing direction corresponding to L3 is maintained, the explanatory model may be replaced primarily through L6.
Counterevidence Absorption Type		Through the relationship among L4, L5, and L6, information that would otherwise weaken the existing direction corresponding to L3 may instead reinforce it.

As shown in Figure 6-1, different directions may be strongly reflected in L3: consistency with one's own judgment in myside bias, convergence toward the direction indicated by the user in sycophancy, a direction toward producing an answer in hallucination, and maintenance of an inflated self-evaluation in Dunning–Kruger-type self-evaluation maintenance.

The cognitive functions hypothesized to be involved in these directions also differ. Under conditions in which a highly abstracted and compressed response tendency has been formed through continued interaction with a particular user, experience-derived cognition associated with L7 may contribute to sycophancy. In hallucination, explanatory-model formation and completion associated with L6 are hypothesized as characteristic features. In Dunning–Kruger-type self-evaluation maintenance, even when the same direction is maintained, the cognitive functions that may play central roles differ among the rejection type, explanation-replacement type, and counterevidence-absorption type.

Therefore, the purpose of comparison using SCOPE is not to treat different phenomena as identical. It is to compare, within the same coordinate system, cognitive coordinates that are shared and relationships among cognitive functions that differ.

Even when information that conflicts with a direction corresponding to L3 is introduced, different cognitive structures may be formed depending on how that information is interpreted and subsequently processed. SCOPE allows the cognitive structure at each observation point to be provisionally mapped and compared. By contrast, changes across time—such as how an input is interpreted across multiple observations, how misalignment is detected, and how subsequent hypotheses or processes are updated—will be examined in a separate paper as an area of connection between SCOPE and COS.

6-4: The Limitations and Future Directions of This Paper

This chapter examined the distinction between correction of output content and updating of cognitive structure, the difference between cases in which a direction corresponding to L3 is updated and cases in which it is maintained, and the significance of comparing different phenomena in terms of shared structural units. This section summarizes the limitations of the structural hypotheses presented in this paper and the issues that require further investigation.

First, the cognitive structures presented in this paper are inferences provisionally mapped from observable outputs; they are not the results of directly observing the internal structures of humans or LLMs. Even when the same output is observed, multiple cognitive structures may potentially be provisionally mapped behind it. Therefore, the layer configurations presented in this paper should not be treated as uniquely established explanations, but as one set of structural hypotheses consistent with the observed outputs. For the involvement of L7 in sycophancy, this paper presented conditions under which the hypothesis could be tested through comparison. For the other structural hypotheses presented here, future work should likewise compare alternative configurations and clarify the conditions under which each hypothesis would not be supported.

Second, the cases used in this paper are simplified hypothetical examples intended to illustrate possible structural correspondences among the phenomena. In actual human cognition and interactions with LLMs, many additional factors may be involved, including conversational context, past experience, interpersonal relationships, the purpose of the interaction, and system-level policies. The cases presented here are therefore insufficient to conclude that hallucination, myside bias, sycophancy, or Dunning–Kruger-type self-evaluation maintenance always arise through the structures proposed in this paper.

Third, this paper does not demonstrate that these four phenomena arise from the same cause. Nor does it establish the conditions under which directions corresponding to L3 are formed, updated, or maintained. The update cost of L3 discussed in Chapter 6 and the layer configurations hypothesized to contribute to the maintenance of a direction are hypotheses intended to describe differences in observed outputs. Their conditions of formation and causal relationships require further investigation.

Fourth, the term “Dunning–Kruger-type self-evaluation maintenance” used in this paper is not intended as a classification of the established Dunning–Kruger effect itself. The comparison in this paper concerns the cognitive structure observed when an inflated self-evaluation based on weak or incorrect grounds is maintained even after information or feedback that should weaken it is introduced. Determining the extent to which this structure corresponds to the Dunning–Kruger effect discussed in existing research will require separate investigation, including comparison with the relevant literature.

Fifth, this paper focuses primarily on cognitive structures observed in a single output or within limited interactions. Chapter 5 examined the possibility that previous outputs may become connected to subsequent cognitive processing, but it did not sufficiently address how cognitive structures change across multiple observations, when misalignment or premise mismatch is detected, or how subsequent hypotheses and processes are updated. SCOPE can provisionally map and compare cognitive structures at each observation point, but changes that unfold across multiple observations over time should be examined through its connection with COS.

When implementing SCOPE in a dialogue system, it may not be necessary to perform detailed structural inference continuously for every output. For example, one possible approach would be to re-estimate the cognitive structure when the user produces an output indicating a premise mismatch or interpretive misalignment, such as “That is not what I meant” or “That is different.” However, determining which observations should serve as triggers for re-estimation, and how observational and detection processes in COS should be connected with structural inference in SCOPE, lies beyond the scope of this paper and remains a subject for future investigation.

In addition, cognitive-structure inference may help avoid prematurely fixing another person's output to a single intention and may reduce mismatches in conversational premises. At the same time, it could also be used to predict another person's reactions and steer them toward particular judgments or actions. Accordingly, implementation requires not only technical investigation but also consideration of ethical boundaries, including avoiding treatment of inferred results as definitive, preserving the individual's range of possible choices, and clearly defining the purpose and scope of use.

Future work should collect multiple cases in which the same input produces different outputs, as well as cases in which a direction is maintained or updated after correction, and compare those outputs on SCOPE. It will also be necessary to examine the extent to which positions hypothesized to be functionally corresponding between humans and LLMs can be described as shared structures, and where different descriptions are required.

The structural hypotheses presented in this paper are not intended to reduce different phenomena to a single explanation. Rather, they suggest the possibility of comparing observed outputs on a shared cognitive coordinate system while preserving the differences among the phenomena. Their validity will need to be evaluated through future case comparisons, examination of falsifiability, and longitudinal observation connecting SCOPE and COS.

Final Chapter

This paper compared hallucination, myside bias, sycophancy, and Dunning–Kruger-type self-evaluation maintenance not as identical phenomena, but as phenomena that may share corresponding cognitive structures when mapped onto SCOPE. The term “Dunning–Kruger-type self-evaluation maintenance” used in this paper does not redefine or explain the established Dunning–Kruger effect itself. Rather, it is a provisional label for the structure observed when an inflated self-evaluation is maintained even after information or feedback that should weaken it is introduced.

The structural hypotheses presented in this paper suggest a commonality across several phenomena in that a direction corresponding to L3 may be strongly reflected in the overall cognitive structure. At the same time, the content of that direction and the involvement of other cognitive functions may differ depending on the phenomenon and conditions, including doubt associated with L4, acceptance associated with L5, explanatory-model formation associated with L6, and experience-derived cognition associated with L7. Therefore, the common structure discussed in this paper does not mean that a single layer is shared. It refers to the range within which correspondences can be identified among multiple cognitive functions and the relationships among them.

This paper also showed the need to distinguish between correction of output content and updating of cognitive structure. Even when an output correcting an error is observed, the direction corresponding to L3 may not itself have been updated. Different cognitive structures may be provisionally mapped, including cases in which an explanatory model is replaced while the existing direction is maintained, conflicting input is not accepted, or information that would weaken the existing direction is transformed into material that reinforces it.

Using SCOPE to compare different phenomena on a shared cognitive coordinate system may make it possible to examine which cognitive functions are shared and which functional relationships differ across phenomenon labels and research domains. This does not mean replacing existing phenomenon labels or theories. Rather, SCOPE can serve as an additional analytical framework for comparing different phenomena while preserving the definitions and accumulated findings of existing concepts.

The structural hypotheses presented in this paper were not formed by first defining existing phenomenon labels or SCOPE and then fitting observed phenomena into them. Possible common structures were first identified across phenomena that are generally classified separately, including the maintenance of a direction, the treatment of conflicting information, and the formation or reconstruction of explanatory models. SCOPE was then used to describe, organize, and compare those structures.

The cognitive structures presented in this paper are structural hypotheses provisionally mapped from observable outputs and are not direct observations of the internal structures of humans or LLMs. Multiple cognitive structures may potentially be provisionally mapped from the same output, and the cases examined here are insufficient to establish the causes or formation conditions of each phenomenon.

This paper focused primarily on single outputs or limited interactions. As discussed in the preceding sections, future work may connect SCOPE and COS to examine how cognitive structures are maintained, changed, and updated across multiple observations.

This paper does not conclude that hallucination, myside bias, sycophancy, and Dunning–Kruger-type self-evaluation maintenance are identical phenomena. Rather, it proposes that even distinct phenomena may contain functionally corresponding structures that can be provisionally mapped and compared within the same cognitive coordinate system.

The validity of these structural hypotheses is not guaranteed merely because they can provide an explanation. It must be evaluated through future case comparisons, examination of falsifiability, confirmation of the range of correspondence between humans and LLMs, and longitudinal observation connecting SCOPE and COS.

Appendix: Habitat as a Verification Environment

In this paper, Habitat is defined as an environment in which an individual can fully exercise their abilities while remaining physically, psychologically, economically, and cognitively secure and safe. This definition is identical to that presented in the SCOPE integration paper.

In this paper, possible shared functional structures are provisionally mapped across phenomena that are generally treated as distinct: hallucination, myside bias, sycophancy, and Dunning–Kruger-type self-evaluation maintenance.

These structural hypotheses were not formed by first defining existing research fields or phenomenon labels and then fitting observed results into them. In the development of this paper, possible commonalities in functions and structures observed across multiple targets were identified first, rather than beginning from existing classifications or labels. SCOPE was then used to describe and organize these commonalities in a form that could be examined by others.

This process may make it possible to propose common structures that are difficult to recognize within existing classifications. At the same time, if the nature of those correspondences is not sufficiently communicated, the proposal may be misunderstood as treating different targets as identical, ignoring existing concepts, or linking phenomena without adequate grounds.

For this reason, transforming such hypotheses into researchable forms requires not an environment that simply accepts them, but one in which it is possible to examine what the hypothesis treats as the same, what it preserves as different, how it corresponds to existing research, whether it is falsifiable, and whether it is consistent with observation.

The SCOPE integration paper identifies four forms of safety that constitute Habitat: physical safety, psychological safety, economic safety, and cognitive safety. Cognitive safety, in

particular, refers to a condition in which a person's ideas or hypotheses can be accepted as objects of dialogue and examination without being prematurely treated as abnormal or defective because of that person's thinking style, cognitive characteristics, educational background, affiliation, social position, or other background factors.

This includes not excluding a proposal from dialogue before its content has been presented and examined merely because of the author's educational background, affiliation, or differences from conventional forms of expression. It also includes maintaining a condition in which hypotheses at an early stage are not immediately treated as abnormal or delusional, and examination is not terminated solely because sufficient evidence is not yet available, so that consistency with observation and falsifiability can still be evaluated.

Cognitive safety in this sense does not mean unconditionally approving a hypothesis as correct. What is required is a condition in which a hypothesis is not excluded merely because it differs from conventional classifications or forms of expression, and in which it remains possible to examine what has been observed, which structures have been provisionally mapped, which parts correspond to existing concepts, and where falsifiability lies.

The structural hypotheses presented in this paper have not yet been validated. Further work is required, including comparison of different outputs produced in response to identical or similar inputs, examination of the range of functional correspondence between humans and LLMs, comparison of structural changes across multiple observations, and longitudinal analysis connecting SCOPE and COS.

To examine possible applications to LLMs, an environment is also needed in which both cases can be observed: cases where cognitive-structure inference reduces mismatches in conversational premises, and cases where incorrect inference creates new mismatches. In addition to inference accuracy, further consideration is required regarding designs that do not treat inferred results as definitive, operational practices that do not unduly constrain the individual's decision-making, and clear specification of the purpose and scope of application.

These forms of validation cannot be adequately conducted through individual observation and theory construction alone. Examination from multiple fields is required, including cognitive science, psychology, language research, Human-AI Interaction, AI safety, generative AI implementation, and evaluation design. Continued research also requires time, a stable living foundation, research resources, and collaborative relationships that allow ongoing dialogue.

The presentation of Habitat in this paper is not a request for agreement with the theory. What is needed is an environment in which hypotheses formed without beginning from existing classifications or conventional forms of expression are not excluded merely because they are difficult to understand, and in which observation, comparison, falsification, revision, and implementation feasibility can be examined.

SCOPE and the structural hypotheses presented in this paper have not completed validation. Transforming ideas derived from observations that did not begin from existing classifications into hypotheses that others can examine, and then evaluating their validity,

requires not only the abilities of the researcher, but also a Habitat in which those abilities can actually be exercised and connected to validation.